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2005 Mathematical Contest in Modeling (MCM) Summary Sheet

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

This paper aims to predict the depletion of crude oil reserve. In our prediction, we analyze the relationship between production, reserve and some exogenous factors in detail. Analyzing the data from website of Organization of Petroleum Exporting Countries (OPEC) and U.S. Energy Information Administration (EIA), we propose three models to forecast the annual petroleum production and peak production years.

Our first model is a short-term prediction for regional annual production. Based on the Hubbert model, we obtain a differential equation for the cumulative production. Using the least squares fits, we obtain the values of parameters for the differential equation. Solving the equation, we get a rough prediction of the oil production.

The second one is used for long-term forecast. Firstly, we model the accumulative discovery by logistic regression; then we introduce the reserve-to-production ratio concept to describe the relationship between the reserve and production; finally, we design a recursion of time to predict the production year by year. According to the variation of reserve-to-production ratio, we consider two scenarios and find the peak year of production is 2010 and 2015 respectively and even till the year of 2100, the annual production can be at the level of 5.5 billion barrels.

The third model takes some exogenous factors impacting production into account. We divide those factors into three categories: Supply factors, Demand factors and Modulation factors. Choosing four main factors, we figured out how they impact production by Neural Network method. Then we check the trend of production variation and the peak year of this model, and find that our prediction agrees closely with Jean Lahrrere's prediction as well as the forecast of the second model.

Basing on our model, we propose recommendations to ensure the sustainable usage and security of oil resource. Finally, we design policies to control the environmental effects and to encourage the new energy and energy-efficiency research.

The future of “black gold”

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Introduction

Petroleum, or crude oil, is a vital exhaustible resource found trapped in certain rocks below the earth's crust. It is a dark, sticky liquid which, scientifically speaking, is classed as a hydrocarbon. Crude oil is highly flammable and can be burned to create energy. Along with its sister hydrocarbon, natural gas, crude oil makes an excellent fuel.

Being viewed as the lifeline of the modern industrial civilization, will petroleum production keep up with the increasing demand? To answer this question, a detailed analysis on production and reserves of petroleum is needed. Petroleum production is deeply influenced by exogenous factors such as global economy, politics, population and environment. So we should take these factors into account. Our goal is to predict the depletion—decline in production—of petroleum through our analysis, and find the connections between those exogenous factors and petroleum depletion. With our prediction, we can answer the question: whether petroleum will be exhausted in the future, or we needn't worry about it.

Symbols Used

P	The annual production of petroleum
A	The cumulative production of petroleum
R	The reserves of petroleum
B_j	Accumulative discovery till the j-th year
D_j	Annual discovery of the j-th year
R_p	The ratio of R/P
bn	Billion (10^9)
b	Barrel

Review of Literature

1 OPEC

Organization of Petroleum Exporting Countries (OPEC) is an international organization of eleven developing countries, which are heavily reliant on oil revenues as their main source of income. The current members are Algeria, Indonesia, Iran, Iraq, Kuwait, Libya, Nigeria, Qatar, Saudi Arabia, the United Arab Emirates and Venezuela.

OPEC's eleven member collectively supply about 40 percent of the world's oil output, and possess more than three-quarters of the world's total proven crude oil reserves.

2 The Energy Information Administration (EIA)

The Energy Information Administration (EIA), created by U.S. Congress in 1977, is a statistical agency of the U.S. Department of Energy. They provide policy-independent data, forecasts, and analyses to promote sound policy making, efficient markets, and public understanding regarding energy and its interaction with the economy and the environment.

3 Hubbert Peak

Hubbert's Peak is the name sometimes given to the highest point on the curve of an Oil Production-Time graph for a country, region or the world. It is named after Dr. M King Hubbert, the geologist who in 1956 predicted the oil production peak of the USA in 1971 (his actual prediction was the mid-1970s). Others have attempted to do the same for the whole world, especially since the late 1980s. The consensus is that it will happen by the end of this decade. In fact, it could have occurred already, in 2000 or 2003, or it could be occurring right now, in 2004.

4 The Logistic Equation

The logistic growth model assumes that a population $N(t)$ of individuals, cells, or inanimate objects grows or diffuses at an exponential rate until the approach of a limit or capacity slows the growth. This model can be expressed mathematically by the following ordinary differential equation, which specifies

the growth rate $\frac{dN(t)}{dt}$ is proportional to the product of $N(t)$ and $M - N(t)$, where M is the upper bound $N(t)$:

$$\frac{dN(t)}{dt} = k(M - N(t))N(t)$$

5 Neural Network

Neural networks are designed to work like a nerve cell system, similar to the workings of the human or biological brain in its physical form. Artificial neural networks are powerful tools for use in classification, empirical modeling, prediction and pattern recognition. Neural networks will never replace conventional methods in some tasks; but as a growing list of applications, the neural architecture will provide either an alternative or a complement to these other techniques.

The sources of data

Given that OPEC's members possess over 75% of world's total proven crude oil reserves, we consider the data on petroleum of OPEC as worldwide. All of this kind of data comes from "OPEC annual statistical bulletin 2003". However, all eleven members of OPEC are developing countries; consequently their gross domestic product (GDP) amount, population and emissions of greenhouse gases cannot represent the whole world's level. So the data on above factors are of the whole world, which comes from EIA.

Methodology

Petroleum production is deeply influenced by many exogenous factors, such as economy, politics, environment and population. For example, the Gulf War, which broke out in 1991, decreases Iraqi and Kuwaiti 1991's annual petroleum production by 86.6% and 77.9% respectively [Annual Statistical Bulletin, OPEC, 2003]. Petroleum production is also affected by the estimates of reserves.

To predict the production and reserves of petroleum, we must examine those exogenous factors. Here we list some major exogenous factors as follows:

- **Time**
Petroleum is a nonrenewable resource, so the production of petroleum can stay the same all the time. It will vary with time.
- **Reserves**
The amount of reserves limits will limits the production of petroleum to keep sustainable.
- **Population**
Petroleum is an essential of our daily life. For example, our car cannot move with petrol. So population will affect petroleum production without doubt.
- **GDP**
The growth of economy depends on energy badly. So GDP can to some extent reflect the demand for energy. A large proportion of energy comes from petroleum. Therefore, GDP is a important factor which influence
- **Emission of CO₂**
CO₂ is a kind of greenhouse gas, the emission of it is concerned by the whole world. The governments will limit the use of petroleum to control the emission of CO₂

Combining all exogenous factors, we can figure out the relationship between those factors and production:

$$P = f(t, R, POP, G, E, L)$$

Where P denotes the annual production; t , R , POP , G , and E denote time, reserves, population, GDP, and emission of CO₂ respectively.

To facilitate representation of the data and a smoother analysis, we firstly find out the influence of one or two factors, then take other factors into account. We divide our prediction into 3 models:

1. Firstly, we just consider the time factor with all the other factors remaining unchanged. $P = f(t)$.
We can use this model to predict the short-term local production.
2. Secondly, we take the change of reserves into account. So $P = f(t, R)$. We use this model to predict the long-term production with respect to the expansion of reserves.
3. Thirdly, we take other factors into consideration. Hence,

$$P = f(t, R, POP, G, E, L)$$

This model is used in the long-term and global prediction. And we can compare the results of it and the second one.

Model 1 Short-term Prediction

1. Dealing with the data

Assuming that all factors remain unchanged except time in this period, we need data have the corresponding feature—stable. We find that the data of all OPEC or the world isn't very stable due to the large area of them. So we select the data of a local area. Saudi Arabia has the largest reserves of petroleum in the world (25% of the proved reserves) [www.cia.gov]. Saudi Arabia also ranks as the largest petroleum exporter around the world, and plays a leading role in OPEC. So Saudi Arabian petroleum production can roughly represent the whole world. Moreover, Saudi Arabia has a relative stable domestic environment. Therefore, we choose the data of Saudi Arabia to create the model. (Detailed data refer to the Appendix A).

In 1956 Hubbert predicted the production of petroleum as a function of time, so we can base on his method to propose a short-term prediction model.

2. Hubbert model

In 1956, M. King Hubbert predicted that U.S. oil production would peak in the early 1970's. Although Hubbert was widely criticized by some oil experts and economists, in 1971 Hubbert's prediction came true [J.H. Laherrère 2000].

Hubbert used the logistic curve to model oil cumulative production:

$$A = \frac{R}{1 + \exp[-b(t - t_m)]}$$

Where A is the cumulative production of petroleum;

R is the ultimate reserve;

b is a factor describing the slope;

t_m is the date at midpoint.

The Hubbert model predicts the production of petroleum in the 1970s accurately. But after 1980 it's prediction went wrong. To predict the production of petroleum nowadays, we need to improve hubbert's method.

3. Improvement of Hubbert model

We know that the logistic equation is given as:

$$\frac{dN}{dt} = r(t)N$$

Where $r(t)$ is proportional to the product of $N(t)$ and $M - N(t)$, with M being the upper bound of $N(t)$.

Cause Hubbert used the logistic curve, we can apply the equation above to predict cumulative production of petroleum

$$\frac{dA}{dt} = r(t)A \quad (1)$$

Here $r(t)$ represents the growth rate of A . To refine Hubbert's method, we need to determine a $r(t)$ more appropriate than logistic model.

Considering that the annual production $P = \frac{dA}{dt}$, we can rewrite the equation as $\frac{P}{A} = r(t)$. To determine $r(t)$, we only need to find the P/A ratio. The relationship between P/A and time is shown in Figure 1

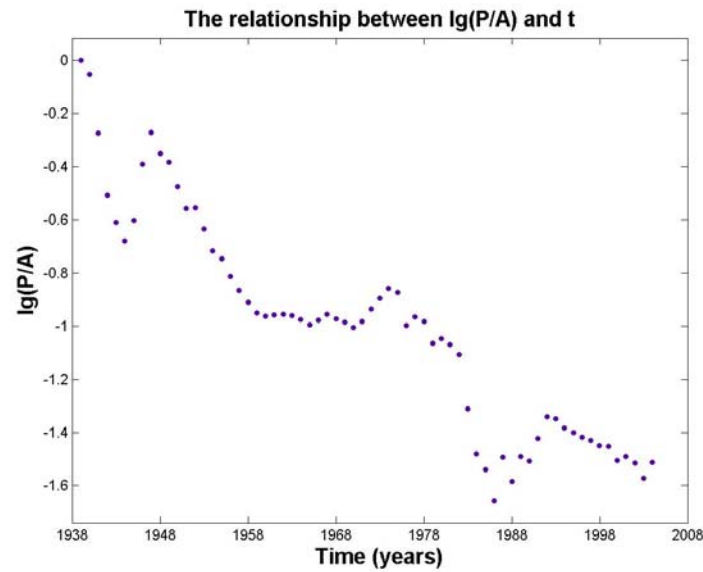


Figure. 1 relationship between P/A and time.

From Figure 1 we can see that the relationship between $\lg(P/A)$ and time is roughly liner, so we generate a liner function for $\lg(P/A)$ based on least-squares fits:

$$\lg(P/A) = -0.0198 - 0.3383t$$

The fitted function and P/A data is shown together in Figure 2:

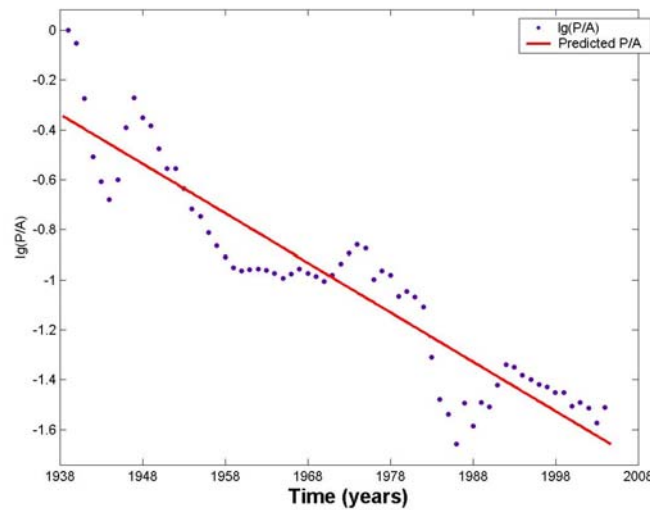


Figure 2 P/A data and fitted function

Figure 2 shows that the P/A data distribute evenly around the fit curve. Some points are far from the fitted curve, which indicates a sudden change in annual production. However, the sudden decline or rising resumes quickly to the fitted function.

Now the P/A ratio is available, or say $r(t)$ is available. To solve equation (1) we need to find a boundary condition. We assume that the petroleum reserve is finite constant, so the cumulative production has an upper bond A_R . So we get the boundary condition:

$$\lim_{t \rightarrow +\infty} A(t) = A_R$$

So equation (1) can be rewrite as follows:

$$\begin{cases} \frac{dA}{dt} = r(t)A \\ \lim_{t \rightarrow +\infty} A(t) = A_R \end{cases}$$

Solving this equation, we obtain that:

$$A = A_R \exp[-10.0604 \exp(-0.0456t)] \quad (2)$$

and:

$$P = \frac{dA}{dt} = -0.3383 A_R \exp[-10.0604 \exp(-0.0456t) - 0.0456t] \quad (3)$$

Where P denotes the annual petroleum production. To determine A and P , we need estimate the upper bound A_R .

We logarithmize both sides of equation (2), and we obtain that

$$\log A = \alpha - \beta x$$

Where $\alpha = \log A_R$ and $x = \exp(-0.0456t)$. Using a least squares fitting, we obtain that $\alpha = 8.303$. Hence we obtain that

$$A_R = 2.0091 \times 10^8 (10^3 \text{ } b) \text{ (} b \text{ means barrel(s))}$$

Now all the parameters to calculate A and P are available.

4. Evaluate

To evaluate the precision of our prediction, we depict the original data and the prediction curve together as follows:

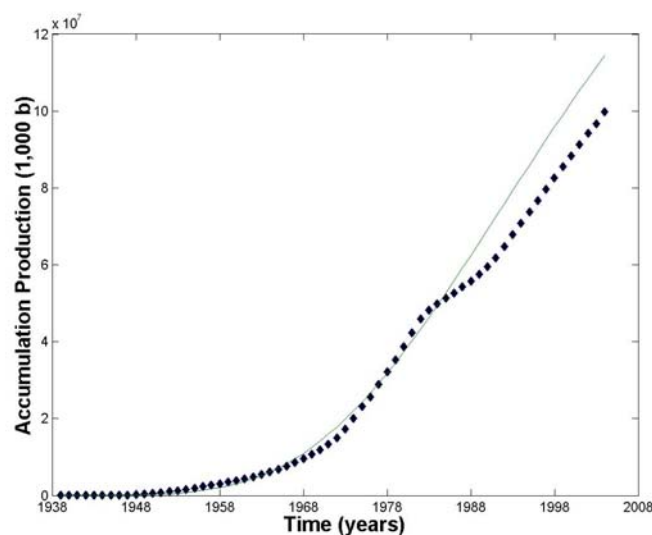


Figure 3 The Saudi Arabian accumulative production data and fitted function

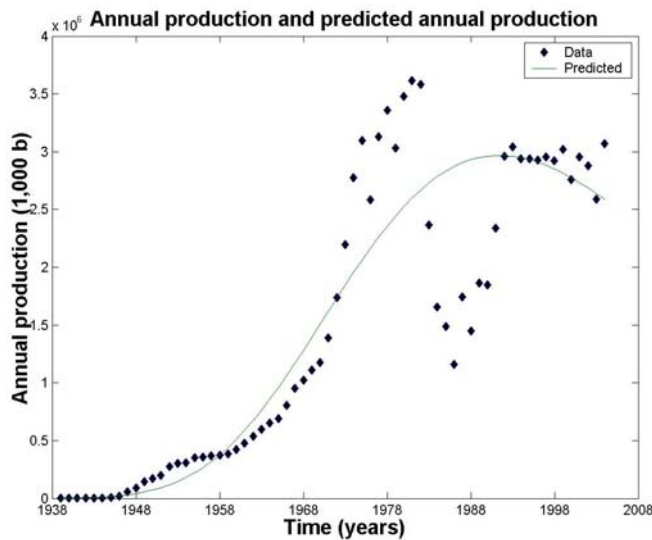


Figure 4 The Saudi Arabian annual production data and fitted function

Form Figure 3 and 4 we can see that our prediction is accurate during the first a few years, but the correctness is decline as time increased.

5. Results of Prediction

We have the predicted annual petroleum production in Saudi Arabia as follows:

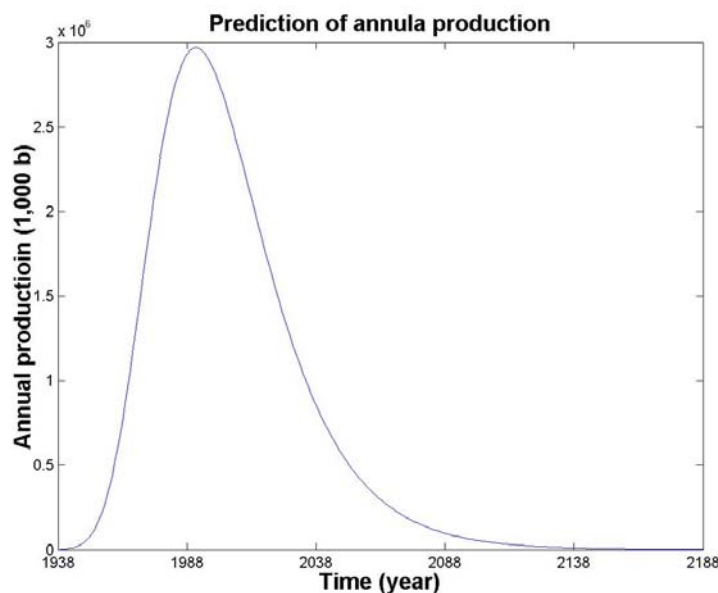


Figure 5 Prediction if annual production

Form Figure 5 we can see before the production peak (around 1990), petroleum production increases exponentially, and then the production will shrink dramatically until about 2150. After 2150, there is nearly no petroleum produced in Saudi Arabia.

Finally, we depict the prediction of Saudi Arabian petroleum reserve as follows:

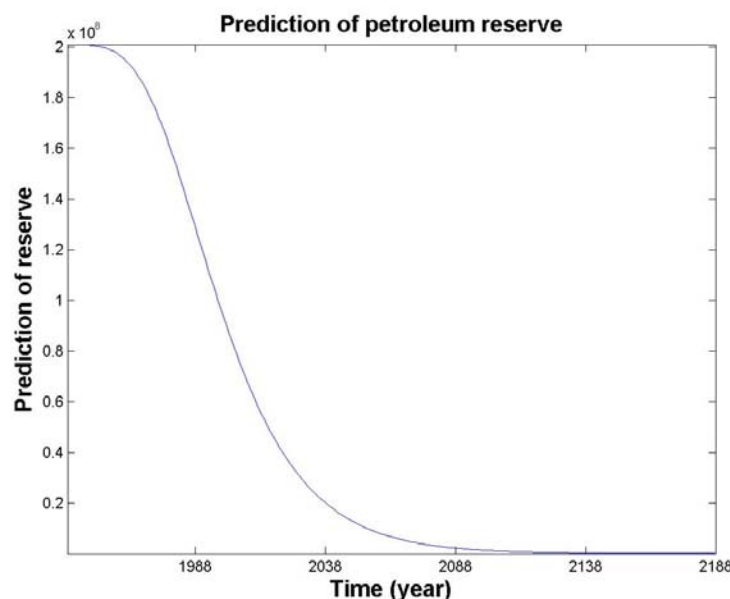


Figure 6 Prediction of petroleum reserves

Figure 6 shows that petroleum reserve will shrink dramatically. After 2100 petroleum reserve in Saudi Arabia will be exhausted.

6. Discussion

Model 1 based on Hubbert's model, which is a mature model in predicting resource production. The result of model is also easy to obtain.

However, model 1 ignores the expansion potential of petroleum reserves. The petroleum reserve is expanding every year. So the long-term prediction of model 1 is not accurate. To predict the long-term trend, we need another model, which take the change of reserve into consideration.

Model 2: A Dynamic Model

To model the production of crude oil over a long time, we need to take the crude oil discovery into account.

1. Discovery data

We do not get any direct discovery data, however, we have the data of OPEC accumulative production and reserves, which is shown in the following table.

Table1
OPEC accumulative production (A) and proven crude oil reserves (R)

Year	1982	1983	1984	1985	1986	1987	1988	1989
A (bn b)	208.61	214.67	220.51	225.95	232.4	238.51	245.4	252.85
R (bn b)	467.37	475.3	510	535.8	643.02	674.02	760.48	764.83
Year	1990	1991	1992	1993	1994	1995	1996	1997
A (bn b)	260.89	269.01	277.74	286.58	295.57	304.55	313.61	322.89
R (bn b)	765.88	771.95	773.7	774.54	777.4	785.07	802.82	805.97
Year	1998	1999	2000	2001	2002	2003		
A (bn b)	333.02	342.59	352.75	362.56	371.43	381.3		
R (bn b)	810.26	818.25	840.54	847.88	881.68	891.12		

Note: 1. Data source: from “Annual Statistical Bulletin, OPEC , 2003.”

2. bn b means billion barrels.

3. A denotes accumulative production.

From the data in this table, we can obtain the OPEC discovery data as follow:

Let the year 1982 be the first year, the accumulative discovery and annual discovery of the j-th year are:

$$B_j = R_j + A_j$$

$$D_{j+1} = B_{j+1} - B_j$$

where

R_j is proven reserves of the j-th year;

A_j is accumulative discovery till the j-th year;

B_j is accumulative discovery till the j-th year;

D_j is annual discovery of the j-th year.

The annual discovery and accumulative discovery we obtained are illustrated in the following figure.

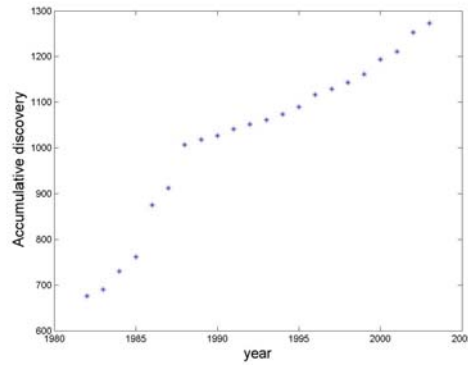


Figure 7. Accumulative discovery.

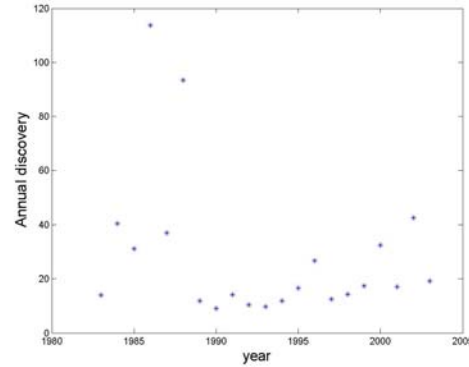


Figure 8. Annual discovery.

From Figure 8 we find that the variety of annual discovery is somewhat irregular, so we only focus on the accumulative discovery.

However, the data in our hands is only for 22 years. It is not enough to reflect the long-term tendency of the accumulative discovery. Hence, we need more analysis.

From figure 7, we see that the increasing rate of accumulative discovery has slow down since the year of 1990. According to the characteristic of logistic model, we can deduce that the increase of B before 1982 is similar to exponential increase, or say:

$$B_i = \alpha \times e^{\beta \times i}$$

Now we need to determine α and β .

To determine α , we need to find an accumulative discover which is long enough before 1982. We can obtain the total accumulative production backdate to the year of 1962 from the OPEC Annual Statistical Bulletin 2003. Finding that the accumulative production and accumulative discovery relate linearly approximately, we multiply the total corresponding accumulative production (A_{1962}) by a coefficient γ to get the accumulative discovery B_{1962} , or say:

$$B_{1962} = A_{1962} \times \gamma$$

To estimate γ , we find the value of B/A for each year from 1982 to 2003 and then get the mean of all the values. In this way, we find $\alpha = 3.5758$ and subsequently find $B_{1962} = 139.4286$, or say: $\alpha = 139.4286$

Now α is available. Hence, we generate:

$$B(i) = 139.4286 \times e^{\beta \times i}, (i = 0, K, 20 \text{ corresponding with the year of 1962 to 1982})$$

Substituting the accumulative discovery of 1982 into the equation above, we get the value of β , which is 0.0789.

Finally, calculating $B(i)$ for all values of i , we generate OPEC accumulative discovery from 1962 to 1981.

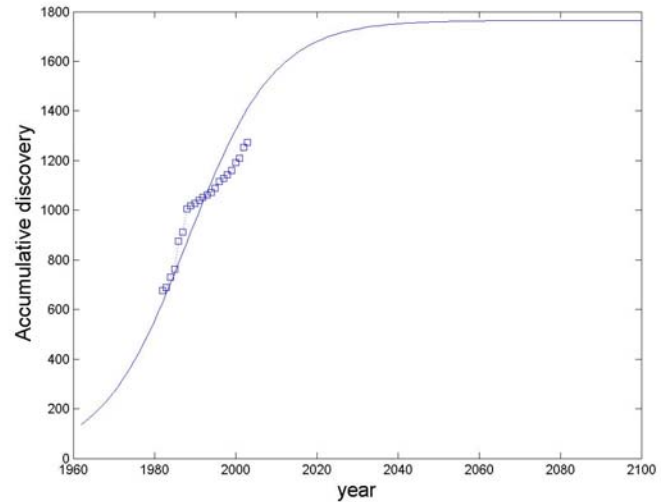
Now, we have accumulative discovery data for the year of 1962 to 2003.

Since the accumulative discovery increases with an upper bound, we consider modeling it by logistic

regression. Let the year of 1962 be the start year, we obtain accumulative discovery by logistic regression:

$$B(t) = \frac{1}{0.006728 \times e^{-0.09406 \times t} + 0.0005666}$$

Now we know how the accumulative discovery changes with time. The variation of accumulative



discovery is illustrated in the following figure.

Figure 9. Variation of accumulative discovery.

From Figure 9, we see that the accumulative discovery reach the ultimate $\frac{1}{5666 \times 10^{-4}} = 1764$ at about the year of 2060.

2. Production

Now we know how accumulative discovery changes with time. To figure out the variation of reserves, we still need to find the production tendency.

Here we introduce the concept of R/P , which means the reserve-to-production ratio. For example, dividing the reserves of 1982 by the production of 1983, we can get the R/P of the year 1983. We use R_p to denote the ratio of R/P

If we could find how the R_p changes as time goes on, then according to $R_p(t)$, we can determine the production of future in the follow way:

1. Make sure a start year; get reserves and accumulative discovery of that year, which is the initial value.
2. According to $R_p(t)$, find the reserve-to-production ratio of the next year; then determine the production of the next year:

$$P(t+1) = \frac{R(t)}{R_p(t+1)} \quad (4)$$

According to the $B(t)$, we can find the accumulation discovery of the next year; then update the reserves:

$$R(t+1) = R(t) + B(t+1) - B(t) - P(t+1) \quad (5)$$

3. If t equals the years we want predict, then stop; else switch into step 2.

The flow chart of it is illustrated as follows:

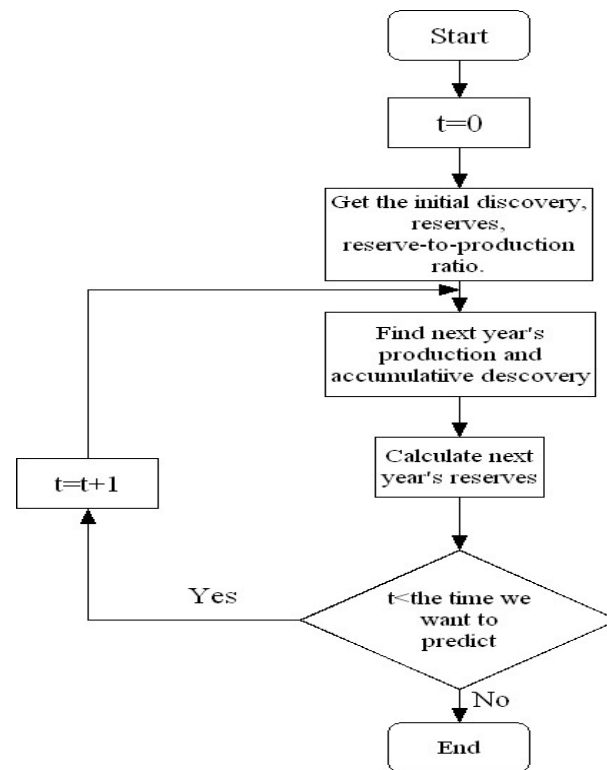


Figure10. The flow chart of determining the production of future.

To use the method above, we need to figure out the variation of $R_p(t)$. Since lots of factors will affect the R_p and we do not have enough data, it is somewhat difficult to determine how it verifies with time.

From the literature, we know that R_p will decrease gradually as time goes on. So, here we give two scenarios:

1. R_p is a constant:

We obtain the constant by calculating the value of $R_p(t)$ for each year and subsequently finding the mean.

2. R_p decreases gradually:

In this scenario, we assume that the initial value of $R_p(t)$ is equal to the constant in the first scenario and the decreasing rate is 1%.

For these two scenarios, the future production is shown in the following figure:

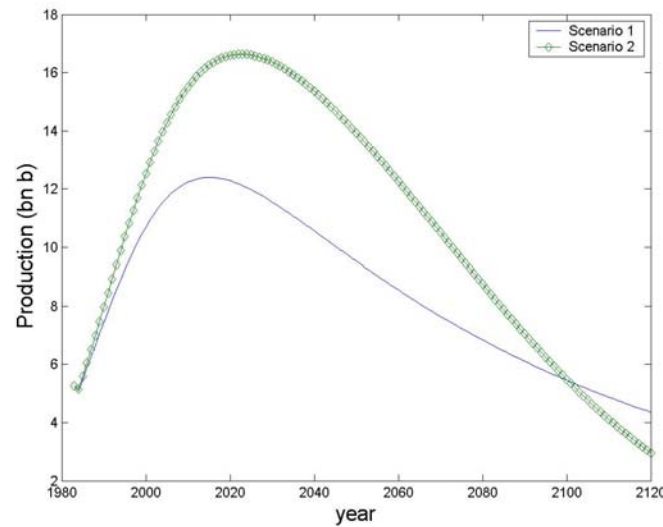


Figure 11. Forecast crude oil production.

From the figure we see that if the R_p is a constant, the peak year of production is about 2010 and the production is about 12.5 billion barrels. Even till the year of 2100, the annual production can be at a level of 5.5 billion barrels. Things become quite different when the R_p decrease at a rate of 1%. The peak year is delayed about 10 years, and the peak annual production can be more than 16 billion barrels.

For both scenarios, we get the depletion of crude oil reserve, which is shown in the following figure.

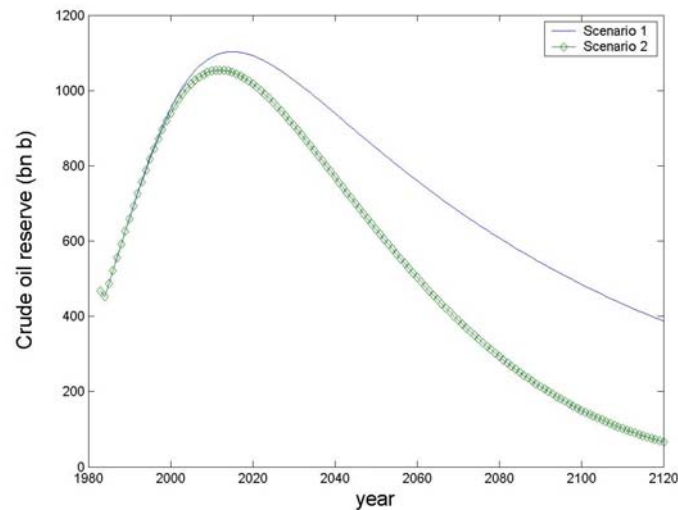


Figure 12. Depletion of crude oil reserve.

Though we deal with is the OPEC data, the situation of the whole world will not be quite different. From both scenarios, we can conclude that the worldwide shortcut of crude oil is not likely to happen.

Using this long-term dynamic model, we can forecast the worldwide depletion of crude oil over a long time. But there are still some drawbacks, such as the lack of accuracy when determine deal with the variation of R_p . To make our model more flexible, we need to take more exogenous factors into account. Hence, adjustment of the model is needed.

Model 3 adjustment of model 2

Since lots of factors will affect the crude oil production and reserve, to account for those factors, we need to analysis the relation between them in detail.

One consideration is to find how those factors affect the R_p . However, when constructing the model 2, we have figured out that the variation of R_p is somewhat irregular. Here, we shift our focus from the R_p to the production. We try to figure out how those factors influence the production directly. If we can predict next year's production, we can use equation (4) and (5) to do the long-term prediction while considering exogenous factors.

We divide all the factors that will affect the production into three kinds as follow:

1. **Supply factor**

Obviously, the reserve is a main factor. The value of the reserve will directly affect the production. Supply factor also includes the capacity of production, which mainly depends on the technology and investment.

2. **Demand factor**

Demand factor mainly concern about the consumption. As the world population and GDP grow, obviously the consumption of oil will increase and subsequently will influence the production.

3. **Modulation factor**

New technology required less energy may appear, and some new energy will play a more important role in the future. This factor in a way can be manifested by some environmental factors such as variation of CO_2 emission:

Besides, politics may influence the production. However, political factor is somewhat different from other factors. Hence, we will consider the politic factor separately instead of including it into a certain kind above.

Base on the analysis above, to adjust our model, we mainly take four factors into consideration, including world population, total GDP of all countries, emission of CO_2 in the world and the crude oil reserve. The value of each factor and annual production from 1983 to 2000 is shown in the following table.

Table 2
The four factors and annual production from 1983 to 2000

Year	Population (Millions)	Total GDP (Billion \$)	Reserve (bn b)	CO_2	Production (bn b)
1983	4653	19613	467.37	5067	6.06
1984	4732	20510	475.30	5241	5.84
1985	4813	21254	510.00	5405	5.44
1986	4905	22026	535.80	5573	6.45
1987	4988	22791	643.02	5702	6.11
1988	5071	23823	674.02	5926	6.89
1989	5161	24727	760.48	6037	7.45
1990	5247	25474	764.83	6126	8.04
1991	5361	26075	765.88	6214	8.12
1992	5444	27260	771.95	6088	8.73
1993	5524	27727	773.70	6093	8.84
1994	5608	28583	774.54	6253	8.99
1995	5688	29376	777.40	6401	8.98
1996	5769	30332	785.07	6553	9.06
1997	5849	31364	802.82	6654	9.28
1998	5928	32032	805.97	6649	10.13

1999	6006	32997	810.26	6492	9.57
2000	6090	34297	818.25	6611	10.16

Note: 1. The value of emission of CO_2 is in million metric tons of C.

2. We obtain the data of population, GDP and CO_2 emission from the website of EIA.

Now, we need to find how the annual production relates with the four factors by using those data, or say we need to find the production function of those four factors. However, we do not know the form of this function. Since the proper neural network can gain upon any kind of nonlinear function, we use the neural network to solve this problem.

Among many kinds of neural network, we choose the Back-Propagated Delta Rule Networks (BP), which is used widely.

BP is a development from the simple Delta rule in which extra hidden layers (layers additional to the input and output layers, not connected externally) are added. The network topology is constrained to be feed forward.

Running BP network consists of:

- Forward pass: the outputs are calculated and the error at the output units calculated.
- Backward pass: The output unit error is used to alter weights on the output units. Then the error at the hidden nodes is calculated (by back-propagating the error at the output units through the weights), and the weights on the hidden nodes altered using these values.

For each data pair to be learned, a forward pass and backwards pass is performed. This is repeated over and over again until the error is at a low enough level (or we give up).

Here we construct a two-layer forward pass BP network with 5 neural cells in the first layer and 1 neural cell in the second layer.

Since those four factors may have some repetitious information, for example, we guess that the population growth is somewhat reflect the increasing emission of CO_2 , so we first standardize the data in table 2, we apply the principal component analysis to the four factors. We find that the principal component is not 4-dimension but 3-dimensions, which proves our guess.

The pretreatment of data and the structure of the two-layer BP network are shown in the following figure.

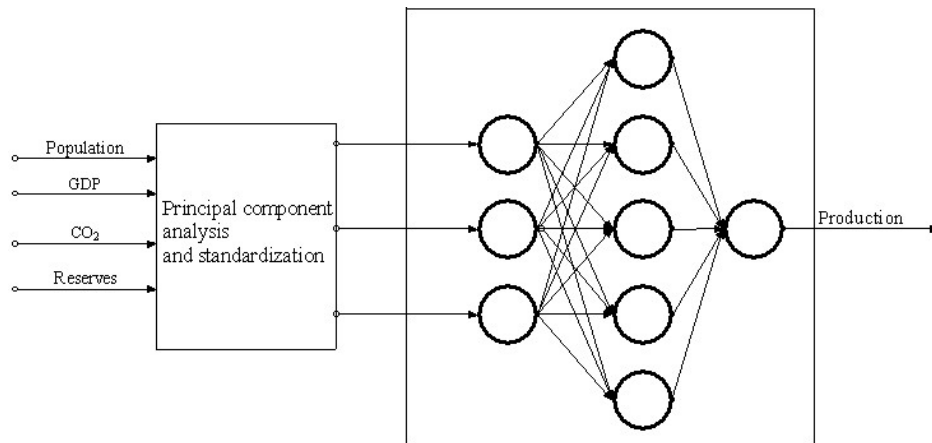


Figure 13. The framework of the two layer BP network

After these disposals, we use the obtained data to train the BP network. The training result is shown in the table 3

Table 3

Training result

Year	Actual production (bn b)	Simulation production (bn b)	Relative error
1983	6.0626	6.06	0.04%
1984	5.8371	5.84	-0.05%
1985	5.4411	5.44	0.02%
1986	6.4493	6.45	-0.01%
1987	6.1093	6.11	-0.01%
1988	6.8883	6.89	-0.02%
1989	7.4605	7.45	0.14%
1990	7.9896	8.04	-0.63%
1991	8.1849	8.12	0.80%
1992	8.7101	8.73	-0.23%
1993	8.8573	8.84	0.20%
1994	8.9857	8.99	-0.05%
1995	9.0177	8.98	0.42%
1996	8.9876	9.06	-0.80%
1997	9.322	9.28	0.45%
1998	10.113	10.13	-0.16%
1999	9.57	9.57	0.00%
2000	6090	34297	818.25

From this table, we see that the relative error between simulation output and actual output is at most $\pm 8\%$. The network we construct has good convergence.

After this training, the network with the trained weight and bias can be used to forecast the production.

Suppose the population function of time is $p(t)$, GDP function of time is $g(t)$ and emission of CO_2 $e(t)$, we derive an algorithm to forecast the future production similar to model 2. The flowchart is as follow:

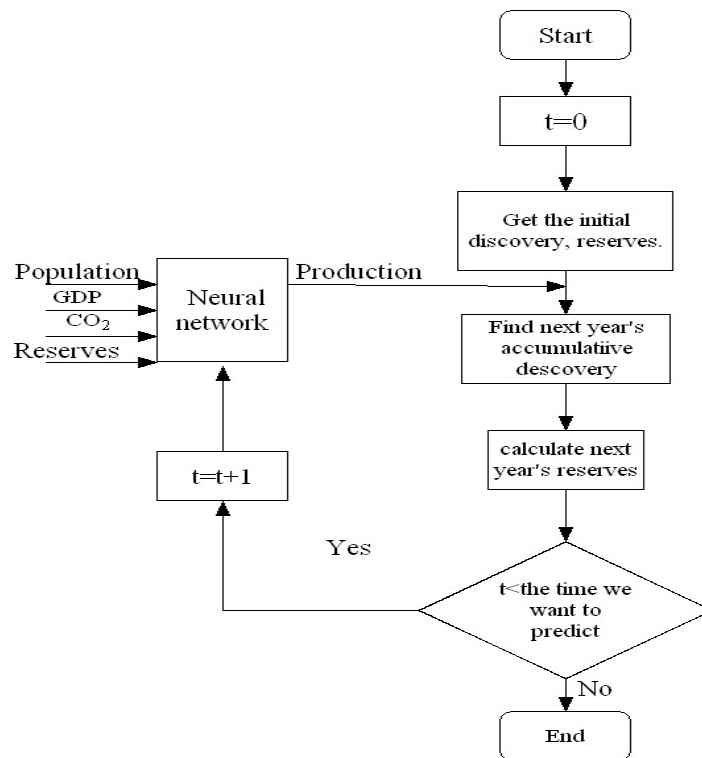


Figure 14. The flowchart of neural network arithmetic

Since the function $p(t)$, $g(t)$ and $e(t)$ all have mature model, here we do not try to figure out the three functions. What we concern about is whether this algorithm can forecast the production correctly.

We check the forecast production of the year from 2000 to 2040 with an interval of 5 years. The value of those four factors we used is cursory, but can reflect the tendency of variation. We use the estimated reserve data of the scenario 1 in model 2, and obtain the other three factors through linear estimate. Then we predict the production of these 8 years:

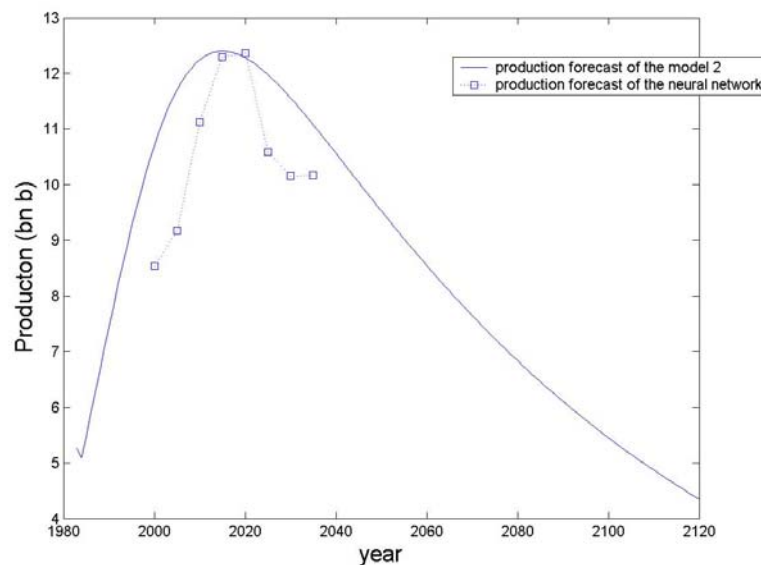


Figure 15. The production forecast of two models

From the figure above, we find that the neural network model have are some discrepancies in the

future production forecast with model 2. However, trend of production variation and the peak year predicted by the network model agree closely with model 2.

Based on the network model, we can figure out the influence of production for each factor to some extent. For example, if the growth rate of GDP increase, the production of oil is likely to be enhanced.

Here we mainly discuss the principle. In fact, using the principle, more factors can be added in this model, which may make the prediction of the oil production and reserve more accurate.

Beside the flexibilities above, there are still some limitations in the neural network model:

1. Some factor such as new technology and new energy is not considered. Since a new technology may greatly deduce the consumption demand for oil, so in the future, technology development and the structure of energy supply may become a important factor.
2. There is still some work to do for this model. We just give the basic principles; more work in detail such as predicting the GDP growth is required.
3. About the politics: Due to the lack of data, we don't consider political factor in our models. The production of petroleum needs a stable political environment. The production will shrink dramatically while the environment become unstable.

Comparison with the prediction of Jean Laherrere:

Jean Laherrere predicts world liquids together with detailed of modeling for conventional oil. The forecast figure is shown as follows: [Jean Laherrere,2001]

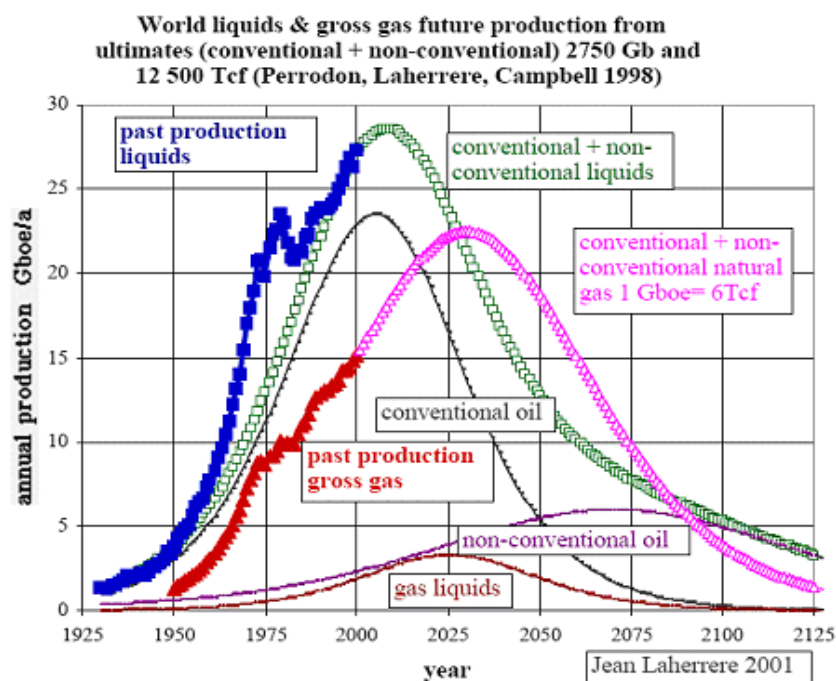


Figure 17. Jean Laherrere forecast figure

From Figure 17 we can see that Jean Laherrere predicts that the production peak of world petroleum will occur in about 2010, which agree closely with our prediction that the production peak will occur between 2010 and 2020 (see figure 11). The quantity of production at the peak in Jean's Laherrere prediction is about 28 billion barrel. Ours are about 12 billion barrel (scenario 1). Consider that we use the data from OPEC, and OPEC supply about 40 percent of the world's oil output, our prediction is very close to Jean Laherrere's.

"Harvesting/management" policy

As we obtained, the oil production will growth with as GDP increases. That is to say as we develop our economy, we consume more resources. To sustain the usage of petroleum during a long period, we should create a series of fair and practical policy. The policy must instruct not only the oil production but also the effectively management of the oil market. We call it harvesting/management policy.

We design the following measures including the economic incentives and disincentives:

1. Economic Incentives:

To keep the consumption remain active while prevent the degradation or rapid exhaustion of the valuable oil, we should work hard to develop the alternative fuel for transportations such as ethanol and solar-powered cars. But there still are many uncertainties and market barriers for new science and technology. So as the instructor the government should make effective measures to enhance the development of energy-efficient and renewable energy technologies. Proposed incentives are summarized below:

1. Invest more money to the development of energy-efficient and renewable energy technologies.
2. Governments should encourage the use and development of energy-efficient and renewable energy technologies.

Tax incentives are thought to be a good idea to reduce renewable energy technologies production costs and create a more mature market for them. Take natural gas from coal seams for example. Natural gas from coal seams production has grown dramatically since the late 1980's in U.S., because of tax credits that provide an incentive for the production of high-cost gas supplies. In 1997, 1,090 billion cubic feet, or 6 percent of total U.S. production, came from coal seams, compared with 41 billion cubic feet in 1988. The tax credit has also contributed to sustained development of natural gas from coal seams by promoting an improved understanding of unconventional gas reservoirs, leading to new and lower-cost technologies for its recovery.

However, the timing, duration, and magnitude of tax incentives should be taking into account.

Other proposed incentives are summarized below:

- Buildings
 - Tax Credits for Energy-Efficient Homes.
 - Tax Credits for Energy-Efficient Equipment in Homes and Buildings.
- Transportation
 - Tax Credits for Electric Vehicles and Fuel Cell Vehicles.
 - Tax Credits for Hybrid Vehicles.
- Renewable Energy Electricity Generation
 - Tax Credits for Wind Generation- Tax Credits for Biomass Generation
 - Tax Credits for Landfill Gas Generation

2. Encouraging international co-operation:

Both producers and consumers will benefit from stable markets that help ensure that supply is sufficient to meet demand and thus contribute to relatively stable global prices. For over a decade, oil and gas producing and consuming countries have been engaged in dialogue on both a bilateral and - through the International Energy Forum on a multilateral basis. The dialogue has helped improve mutual understanding, confidence and awareness of long-term common interests as well as promoting the sustainable development of specific initiatives such as the Oil Data Transparency exercise. As trade in energy increases, such dialogue will become more and more important.

Sustainable energy solutions also have the potential to strengthen energy reliability. More effective work is urgently needed to promote the deployment of renewable sources of energy in developing countries as well as encouraging investment in appropriate energy infrastructure worldwide.

Security of the lifeline

In order to keep the oil resources from misuse and disruption, and unnecessary degradation or destruction of the resource, we recommend the following measures and policies.

- Lay more emphasis on the effective management

The management of the oil exploitation must guarantee the production in a normal and robust state. So the more effectively we apply the management, the more securely we use the oil.

Since the information technology (IT) infrastructure is now applied to many large oil companies' production and even the entire operation [Richard 2004], the production and transportation can realize the remote control.

However, at the same time, threats to IT network security are not only an IT security problem anymore but also a great potential risk to the management of oil production. For example, the offshore platform is connected with the remote control center. If the network were attacked, the data of the daily production may be stolen, and the hacker may even command the platform using the administrative authority. Nevertheless, the worst case is that the whole network may be shut down by the hack attack using some sophisticated virus, then the management system may collapse and the overall production may be interrupted, this would bring about the loss of millions of dollars.

- Prevent the disorderly extraction

The disorderly extraction is a serious problem, which may lead the soil pollution and water pollution and the greenhouse gas emission in large quantities. Lack of advanced equipments and optimization of the production procedures, the disorderly extraction has considerable risk of causing the environment pollution, and change the local ecosystem, so regulation and authorized production is urgently needed in some fields and refineries

- Prevent the Oil stealing

Oil stealing should also be prevented, since many oil theft often occur along the pipe laying, the destructor may damage the transport equipment when stealing oil.

Potential emergency in the oil field may cause the fluctuation of the oil production and the interruptions of the oil supply.

With the development of the technology the our knowledge to the oil resources, following all these policies may not guarantee the solution to the security of the oil utilization, but the pay-off will be worth the effort.

Environmental effect

1. Pollutants

Exploitation and consumption of petroleum may produce many sources of the pollutants, which pose great hazard to human health. Pollutants receiving greatest concern and their sources are:

- Benzene from vehicle emissions and from the handling of petrol, a human carcinogen.
- Butadiene (a hydrocarbon) emitted in car exhausts and a suspected human carcinogen.
- Carbon monoxide emitted from power stations, road transport, combustion plant – low level doses can affect fertility, reflexes and impair thinking, high level doses are fatal.
- Lead emitted from vehicles using leaded petrol and from the incineration of plastic packaging materials – inhalation of lead particles in air damages the nervous system, affects blood pressures and impairs mental development.
- Nitrogen dioxide is produced by power stations and road transport – it contributes directly to the formation of smog and can directly cause respiratory illness and reduced lung function.
- Ozone is formed in photochemical reactions involving carbon monoxide, and nitrogen oxides – it is a strong pungent blue gas that is an irritant to the eyes, nose and throat and can lead to respiratory and other health problems.

- Particulate Matter (PM₁₀) emitted by diesel fuelled road vehicles and other combustion plant producing black smoke or soot – contributes to smog formation, carries carcinogens into organisms, damages plants and buildings
- Sulphur dioxide from combustion of sulphur containing oil and coal, predominantly from power stations – causes breathing difficulties and has synergistic effects with particulates and moisture to form sulphuric acid in the lungs.

(Source from: Briefing paper on environmental issues, UK, NHS Purchasing and Supply Agency, January 2004).

2. Increasing susceptibility of natural disaster

The recent tsunami in the Indian Ocean has received the global attention and a worldwide concern about the natural disasters has been provoked. Researchers said that global warming play a major role in this tragedy. But emissions from vehicles using various fuels cause more and more serious air pollution and mainly contribute to global warming.

In urban areas, transportation is the major source for a number of pollutants (lead, nitrogen dioxide, carbon monoxide, sulphur dioxide, particles, benzene and 1,3-butadiene). CO₂ emissions from traffic also have a global environmental impact. Road transport is the fastest growing source of global emissions of carbon dioxide—the main greenhouse gas. Nitrous oxide and ground level ozone are other greenhouse gases that arise in part from vehicle emissions.

According to the statistics, since 1751 roughly 283 billion tons of carbons have been released to the atmosphere from the consumption of fossil fuels and cement production, oil consumption is doubtlessly the main source. Half of these emissions have occurred since the mid 1970s. The 2000 global, fossil-fuel CO₂ emission estimate, 6611 million metric tons of carbon, represents a 1.8% increase from 1999. The average annual fossil-fuel release for the decade 1990-99 was 6.35 billion tons of carbon [Marland et al 2003]. The global CO₂ emissions from fossil-fuel burning is show as follows:

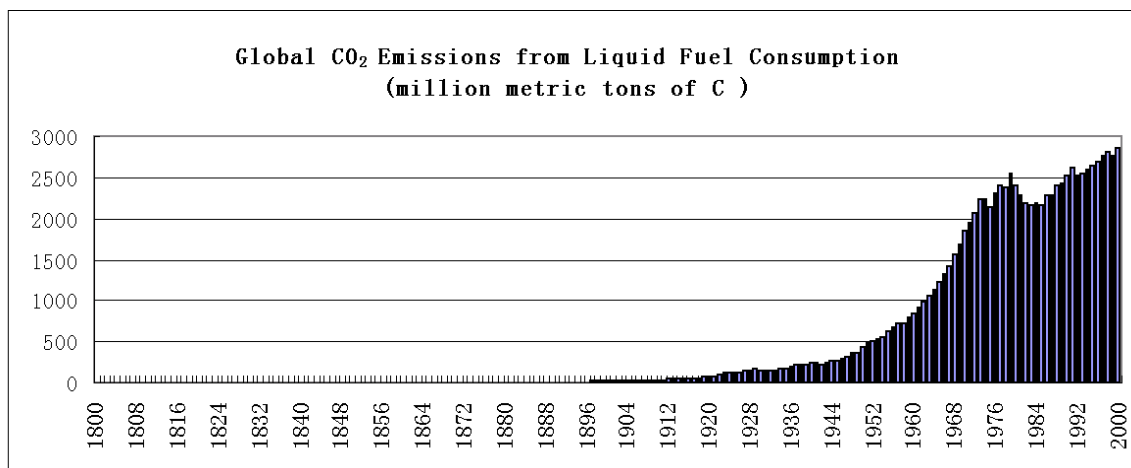


Figure 18. Global CO₂ Emissions from Fossil-Fuel Burning:(1800-2000).

(Data source: Carbon Dioxide Information Analysis Center, Oak Ridge National Laboratory, and University of North Dakota)

From Figure18, we can the emissions from fossil-fuel burning increases dramatically after the World War 2 (1945).

3. Policies to control emission of pollutants and carbon dioxide:

Carefully store the Oil:

Oil must be stored in a way that will not harm the environment. Pollution of the environment from poor waste storage occurs in many ways. Improving the way materials are stored and handled on site doesn't require a great deal of money. Handling and storage of waste on site can be improved using some simple techniques:

Check the refueling area for the following:

- All tanks are bounded.
- There are no signs of damage or corrosion.
- Pipes, valves and gauges are within the bund.
- It is isolated from the drainage system.
- A spill station is located nearby (see Spills module).

When refueling all staff should ensure that:

- Refueling doesn't start until the nozzle is within the vehicle.
- The nozzle isn't removed until the flow of fuel has stopped.
- Any spills that do occur are cleaned up quickly and disposed of correctly.
- Pipes and nozzles are stored correctly within the bund when finished.

4. **Control the use of fossil fuels and encourage the cleaner system**

Uncontrolled use of fossil fuels could have major local, regional and global environmental impacts. Cleaner systems mitigate and even neutralize the adverse consequences of the use of fossil fuels and permit their positive qualities to be more fully harnessed for economic and social development. The technology for these systems is advancing rapidly. A broad range of stakeholders must understand the high value of these systems in order to ensure that cleaner fossil energy systems will be used and that fossil energy will be sustainable.

Find alternatives to Oil

Facing the serious greenhouse effect caused by the vehicles emissions, transportations need new energy-powered vehicles. Nowadays, vehicles using Liquefied Petroleum Gas (LPG) is still on the market. However, LPG is a byproduct of natural gas processing or crude oil refining. It consists mainly of propane (over 90 percent), and thus LPG is usually referred to as propane. It also contains varying amounts of butane, propylene, and butylenes. New alternatives are continually turning up. Other alternative-fueled vehicles which has been used in U.S. includes the following types:

Table 4. Alternative-fueled vehicles types.

LPG	Using Liquefied Petroleum Gas
Non-LPG	Using Compressed Natural Gas (CNG)
	Using Liquefied Natural Gas (LNG)
	Using Methanol, 85 Percent (M85)
	Using Methanol, Neat (M100)
	Using Ethanol, 85 Percent (E85)
	Using Ethanol, 95 Percent (E95)
	Using Electricity

Not only is crude oil the basis of the world's first trillion-dollar industry, it also is the largest item in the balance of payments and exchanges between nations. The world's economy, so far, has been driven by an abundant supply of cheap oil. However, being nonrenewable and exhaustible, oil may eventually be short of demand. Extreme shortage will cause an oil crisis. To get out of this dilemma, we must find new energy.

Where is the new energy? We must resort to science and technology innovations. New Energy utilization is defined as the technology essential for the introduction of alternative energy to petroleum, which is technically well on its way to practical use, but is not widely used due to the restriction of economic aspects. Therefore, for energy technologies that have already reached the practical use stage, such as hydroelectric power generation and geothermal power generation, as well as

technologies that are still in the research-development stage, such as wave power generation and ocean thermal power generation are categorized as natural energy, not new energies.

The new energies consist of the technologies which is shown in the following table:

Table5 Basic information of new energy

New Energy	Basic information
Photovoltaic power generation	Application has increased by 3.5 times during the last 3 years. Generating cost is still high, even though the price of the system has come down over the last six years to one fourth of what it had been.
Wind power generation	Application increased by about 7 times during the last 3 years, and its commercial viability has been recognized depending on local conditions. Future issues are economic efficiency and stability.
Waste power generation	Local communities are mainly promoting the installation of facilities. A future issue is the problem of location, etc.
Biomass power generation	Wooden chips, bagasse (sugar cane husks) and sludge are used mainly. In recent years, there have been trials using methane gas from the food waste, but still economic viability is still an issue.
Solar thermal utilization	Level of application has decreased in recent years. The main issue is economy.
Thermal utilization of biomass	Black liquor waste accounts for a substantial amount of the new Energy in this area, such as biomass heat utilization.
Clean energy motor vehicles	Hybrid automobiles and natural gas automobiles are steadily increasing, and the level of application amounted about 4-fold during the last 3 years. Future issues are economic efficiency, performance and infrastructure availability.
Natural gas cogeneration	The application increased by about 1.4 times during the last 3 years. High efficiency facilities are still expensive.
Fuel cell	Phosphoric acid type has decreased. Competition to develop solid high polymer types is already noticeable in domestic and overseas companies to develop cells for practical use and dissemination. Large-scale application is anticipated in future.

(Source: Agency for Natural resources and Energy, Japan)

These new energies mentioned above are contributing to constructing sustainable economic structure, as they are environmentally friendly, emit little CO₂, and provide us with alternatives for relying less on petroleum. In addition, introduction of new Energy technologies can bring further advantages such as creating new industry and chance of employment.

Natural energy, such as photovoltaic power generation or wind power generation, is inexhaustible, and does not increase CO₂, the cause of global warming.

Research Policy:

We recommend investment in development of new energy technology. Spending on research and development (R&D) has always been associated with the pace and quality of technological innovation in all directions. In the field of energy, it is a widely held view that spending on R&D is an important precursor to the technological advances required to secure sufficient, safe and environmental- friendly energy supplies, and to use them more efficiently.

We need to develop research policy, which encourages the research of a wide range of renewable energy and its utilization, and to make significant changes to our institutions and systems. Developing new energy and energy-efficiency technology is the only way to ensure our cheerful prospect.

- Raise the awareness of the importance of the new energy technology.
- Determine the precise benchmark of energy efficiency performance using case studies on energy efficiency performance for transportation and industrial sectors (vehicles and cement, steel etc.)
- Set up the specific research center, which collect good ideas and experiences on the R&D, provide the totally open source of information, share all of the work, findings and research.
- Increasing the investment in the new-energy and energy efficiency study, ensuring adequate and appropriate funding is essential, especially the cases such as the solar energy and solar-powered vehicles.
- Promoting the co-operation among the universities, colleges and other research institution. Moreover, from a worldwide view, international co-operation should also be encouraged and organize the worldwide communications and exchange
- Encourage the new technologies diffusion.

Ultimately, what we should do is to relentlessly encourage the research and development, furthermore the demonstration of the new energy technologies. The more powerful and environment-friendly energy will be put into use in coming years,

We should have every confidence in our cheerful prospect.

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Appendix A: the average annual production and cumulative production of Saudi Arabia (1,000 barrel(s))

year	Daily average	Cumulative	year	Daily average	Cumulative
1938	1. 4	511	1971	4, 768. 90	15, 024, 497
1939	10. 8	4, 453	1972	6, 016. 30	17, 226, 462
1940	13. 9	9, 540	1973	7, 596. 20	19, 999, 075
1941	11. 8	13, 847	1974	8, 479. 70	23, 094, 166
1942	12. 4	18, 373	1975	7, 075. 40	25, 676, 687
1943	13. 3	23, 228	1976	8, 577. 20	28, 815, 932
1944	21. 3	31, 024	1977	9, 199. 90	32, 173, 898
1945	58. 4	52, 340	1978	8, 301. 10	35, 203, 793
1946	164. 2	112, 273	1979	9, 532. 60	38, 683, 202
1947	246. 2	202, 136	1980	9, 900. 50	42, 306, 785
1948	390. 3	344, 986	1981	9, 808. 00	45, 886, 705
1949	476. 7	518, 981	1982	6, 483. 00	48, 253, 000
1950	546. 7	718, 527	1983	4, 539. 40	49, 909, 884
1951	761. 5	996, 474	1984	4, 079. 10	51, 402, 828
1952	824. 8	1, 298, 351	1985	3, 175. 00	52, 561, 686
1953	844. 6	1, 606, 630	1986	4, 784. 20	54, 307, 919
1954	961. 8	1, 957, 687	1987	3, 975. 20	55, 758, 849
1955	976. 6	2, 314, 146	1988	5, 100. 10	57, 625, 503
1956	1, 002. 80	2, 681, 171	1989	5, 064. 50	59, 474, 046
1957	1, 030. 80	3, 057, 413	1990	6, 412. 50	61, 814, 608
1958	1, 058. 50	3, 443, 765	1991	8, 117. 80	64, 777, 605
1959	1, 152. 70	3, 864, 501	1992	8, 331. 70	67, 827, 007
1960	1, 313. 50	4, 345, 242	1993	8, 047. 70	70, 764, 418
1961	1, 480. 10	4, 885, 478	1994	8, 049. 00	73, 702, 303
1962	1, 642. 90	5, 485, 137	1995	8, 023. 40	76, 630, 844
1963	1, 786. 00	6, 137, 027	1996	8, 102. 30	79, 596, 286
1964	1, 896. 50	6, 831, 146	1997	8, 011. 70	82, 520, 556
1965	2, 205. 30	7, 636, 080	1998	8, 280. 20	85, 542, 829
1966	2, 601. 80	8, 585, 737	1999	7, 564. 70	88, 303, 945
1967	2, 805. 00	9, 609, 562	2000	8, 094. 50	91, 266, 532
1968	3, 042. 90	10, 723, 264	2001	7, 888. 90	94, 145, 980
1969	3, 216. 20	11, 897, 177	2002	7, 093. 10	96, 734, 962

1970	3, 799. 10	13, 283, 848	2003	8, 410. 30	99, 804, 721
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Appendix B: The neutral network model

% the neutral network model

%The data to be trained %%%%%%%%%%%
%%%%%%%%%

```
p=[4653 4732    4813    4905    4988    5071    5161    5247    5361    5444    5524    5608
5688    5769    5849    5928    6006    6090
19613  20510  21254  22026  22791  23823  24727  25474  26075  27260  27727  28583
29376   30332  31364  32032  32997  34297
467.37 475.3   510 535.8   643.02 674.02 760.48 764.83 765.88 771.95 773.7   774.54
777.4   785.07   802.82 805.97 810.26 818.25
5067   5241   5405   5573   5702   5926   6037   6126   6214   6088   6093   6253
6401   6553   6654   6649   6492   6611
];
t=[6.06 5.84    5.44    6.45    6.11    6.89    7.45    8.04    8.12    8.73    8.84    8.99
8.98   9.06   9.28   10.13  9.57   10.16
];
```

%%%%%%%%%

```
[pn,meanp,stdp,tn,meant,stdt]=prestd(p,t);           %standardization
[ptrans,transMat]=prepca(pn,0.001);                 % principle component analysis
[R,Q]=size(ptrans);
```

```
net=newff(minmax(ptrans),[5 1],{'tansig' 'purelin'},'trainlm'); %training the network
```

```
net.trainParam.show=5;
net = init(net);
[net,tr]=train(net,ptrans,tn);
an=sim(net,ptrans);
a=poststd(an,meant,stdt);
```

```
n= 18:5:53;
%%%%%%%%%           % Estimate future production
pop= 85.57.*n+4567; % the cursory estimate of population
g = 750.8.*n+18700; %the cursory estimate of population
e=87.63.*n+5228;    % the cursory estimate of population
testa= zeros(4,8);

%get the reserve data based on the model 2
testa(3,:)= [951.79   1040.4  1088.4  1102.7  1092.2  1065.1  1027.9  984.92];
```

```

testa(1,:) = pop;
testa(2,:) = g;
testa(4,:) = e;
test1=trastd(testa, meanp, stdp);
test2=trapca(test1,transMat);
result1 = sim(net,test2);
result1 = poststd(result1,meant,stdt);          %the result of the neutral network model

```

Appendix C: the long-term dynamic model

```

%the long-term dynamic model
pro1 = zeros(1,138);          %scenario 1
res = 467.3                    % predict the reserve of crude oil since 1983
accP = 208.61;
resArray1 = zeros(1,138);
for pre=21:158
    preDis=1./(0.006728*exp(-0.09406*pre)+0.0005666);
    temp = res/88.9108
    pro1(pre-20) = temp;
    accP = accP + temp;
    resArray1(pre-20) = res;
    res = preDis - accP;
end
pro2 = zeros(1,138);          %scenario 2
res = 467.37;
accP = 208.61;
rprate = 88.9108;
resArray2 = zeros(1,138);
for pre=21:158
    preDis=1./(0.006728*exp(-0.09406*pre)+0.0005666);
    temp = res/rprate;
    rprate = rprate * 0.99;
    pro2(pre-20) = temp;
    accP = accP + temp;
    resArray2(pre-20) = res;
    res = preDis - accP;
end
n=1983:2120;

                                %plot needed figure

plot(n,pro1,n,pro2);
plot(n,resArray1,n, resArray2);
m=2000:5:2035;

```

```
plot(n,pro1,m,result1,'bs');
```